Sangmin Lee

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Profile

An adept learner with analytical thoughts. I have extensive publications in computer science, and artificial intelligence. Currently, I am interested in designing efficient neural networks and brain-inspired AI.

Education and Qualifications

Mar 2021- Jul 2023 **Dongguk University,** Seoul, South Korea
MA in Artificial Intelligence
GPA: 3.81 / 4.5

- **Thesis:** "MPTFL: Computation and Communication Efficient Federated Learning with Multi-prize Lottery Ticket Hypothesis"
  Advisor: Hyeryung Jang, Assistant Professor in Artificial Intelligence, DGU

Apr 2015- Feb 2020 **National Institute for Lifelong Education**, Seoul / South Korea
BA, Department of Computer Engineering
GPA: 4.0 / 4.5

Publications

- **Sangmin Lee**, Hyeryung Jang, "BipruneFL: Computation and Communication Efficient Federated Learning with Binary Quantization and Pruning", *IEEE Access*, 2025
- **Sangmin Lee**, Hyeryung Jang, "Accurate and Computation-efficient Federated Learning with Multi-prize Lottery Ticket Hypothesis.", *Korea Artificial Intelligence Conference, The Korean Institute of Communications and Information Sciences*, 2022
- **Sangmin Lee**, Hyeryung Jang, "A Study on Training Binary Neural Networks using Efficient Gradient Computations for Intelligent Systems", *International Symposium on Advanced Intelligent Systems (ISIS)*, 2021.
- **Sangmin Lee**, Wonho Son, et al. "Face Frontalization Model with A.I. Based on U-Net using Convolutional Neural Network.", *Korea Information Processing Society*, 2020
- JiYun Kim, **Sangmin Lee** et al. "Food Exchange Table Organization Model Based on Decision Tree Using Machine Learning.", *Korea Information Processing Society*, 2020
- **Sangmin Lee**, Jongwan Kim, et al. "Study on the Performance Evaluation and Analysis of Mobile Cache Memory.", *Korean Society of Computer Information*,2020
- Jihyun Kim, **Sangmin Lee** et al. "Nearest Neighbor-based Pre-processing Scheme for Advanced Skyline Query.", *Korea Information Processing Society*, 2020

Work / Research Experience

Feb 2026 – Dec 2026 **COGI LAB**., Department of Brain & Cognitive Sciences**,** KAIST, South Korea

Part-time researcher

- Conduct research on continual learning and biologically inspired neural networks.
- Develop and evaluate learning algorithms to mitigate catastrophic forgetting.
- Manage and maintain GPU computing servers
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Sep 2023 – Feb 2024 **Intelligence and Optimization in Networks LAB**., Dongguk University, South Korea

Researcher, Intern

- MA research to enhance efficiency in federated learning using quantization & pruning methods
- Research paper: "BipruneFL: Computation and Communication Efficient Federated Learning with Binary Quantization and Pruning" to IEEE Access (November 2024)

Dec 2019 – Jan 2021 **Artificial Intelligence / Big Data Software Code LAB**., Sahmyook University, South Korea

Research Student

- Participated in weekly paper seminars to enhance research skills
- Published multiple papers in computer science and machine learning, with two papers receiving awards

Relevant Training

Jun 2019 – Nov 2019 **Bit Academy**, Seoul, South Korea
Participation & Research, Course for Fostering Top AI Platform Developers

- Progressed from building a simple MNIST classification model to developing complex GAN-based image generation models
- Completed a team project on autoencoder-based face frontalization, The project was published and received the Best Research Paper award

Mar 2017 – Sep 2017 **Bit Academy,** Seoul, South Korea
Participation and Research, Course for Fostering ICT Infra Solution Architecture Developer

- Developed proficiency in multiple programming languages, C, C++, C#, Python, Unity
- Team leader in a VR project using Unreal Engine, while also developing several mini-projects, including a C-based mouse-clicking game and a C# messenger application

Awards

Dec 2021 **Awards, Best Presentation for Research**, International Symposium:Advanced Intelligent Systems (ISIS), South Korea
Research Thesis: *"A Study on Training Binary Neural Networks using Efficient Gradient Computations for Intelligent Systems"*

Nov 2020 **Awards, Best Research Paper**, Korea Information Processing Society, South Korea
Research Thesis: *"Face Frontalization Model with A.I. Based on U-Net using Convolutional Neural Network"*

May 2020 **Awards, Research Excellence,** Korea Information Processing Society, South Korea
Research Thesis: *"Nearest Neighbor-based Pre-processing Scheme for Advanced Skyline Query"*

Skills

Certificates: **Certificate of Engineer Information Processing**, Issued by Human Resources Development Service of Korea
Certificate of DELF B1 Level, Issued by France Education International

Languages: **Korean** (Native), **English** (Fluent), **French** (Advanced, DELF B1)